Sturm Comparison Theorem

- **Statement:** Let $y(x)$ and $z(x)$ be nontrivial solutions of

$$y'' + p(x)y = 0, \qquad x \in I,$$

and

$$z'' + q(x)z = 0, \qquad x \in I,$$

where p and q are positive continuous functions and $p > q$ on I.

- Then between any two consecutive zeros x_1 and x_2 of $z(x)$, there exists at least one zero of $y(x)$ unless $p \equiv q$ on (x_1, x_2).
- **Proof:** Consider x_1 and x_2 with $x_1 < x_2$. WLOG, assume $z > 0$ in (x_1, x_2).
- Then $z'(x_1) > 0$ and $z'(x_2) < 0$ (observe from the graph of $z(x)$).
- Suppose on the contrary that y has no zero on (x_1, x_2). Assume $y > 0$ in (x_1, x_2).



Sturm Comparison Theorem (Contd.)

- Let W be the Wronskian of y and z, i.e., $W = yz' - zy'$.
- Then $\dfrac{dW}{dx} = yz'' - zy''$.
- We know $y'' + py = 0 \implies y'' = -py$ and $z'' + qz = 0 \implies z'' = -qz$.
- From above, on (x_1, x_2),
  $\dfrac{dW}{dx} = yz'' - zy'' = y(-qz) - z(-py) = (p - q)yz > 0.$
- Integrating between x_1 and x_2:

$$W(x_2) - W(x_1) = \int_{x_1}^{x_2} (p - q)\, y(x)z(x)\, dx. \tag{18}$$

- Now $W(x_2) = y(x_2)z'(x_2) \leq 0$ and $W(x_1) = y(x_1)z'(x_1) \geq 0$, so LHS of (18) ≤ 0.
- But RHS of (18) is strictly greater than zero unless $p \equiv q$ on (x_1, x_2) – a contradiction.
- This proves that between any two consecutive zeros x_1, x_2 of z, there exists at least one zero of y unless $p \equiv q$ on (x_1, x_2).



Zeros of Bessel's Function

- **Theorem:** The Bessel function of the first kind J_ν $(\nu \geq 0)$ has infinitely many positive zeros.
- **Proof:** The number of zeros of J_ν equals the number of zeros of a nontrivial u satisfying (17):

$$u'' + \left(1 + \frac{1/4 - \nu^2}{x^2}\right) u = 0. \tag{19}$$

- For large enough x, say x_0, we have

$$\left(1 + \frac{1/4 - \nu^2}{x^2}\right) > \frac{1}{4}, \qquad x \in (x_0, \infty). \tag{20}$$

- We compare (19) with

$$v'' + \frac{1}{4} v = 0. \tag{21}$$

- Due to (20), between any two zeros of a nontrivial solution of (21) in (x_0, ∞), there exists at least one zero of a nontrivial solution of (19).



Zeros of Bessel's Function (Contd.)

- $v = \sin(x/2)$ is a nontrivial solution of (21), which has infinitely many zeros in (x_0, ∞).
- Hence, any nontrivial solution of (19) has infinitely many zeros in (x_0, ∞).
- Thus, J_ν has infinitely many zeros in (x_0, ∞), i.e. J_ν has infinitely many positive zeros.
- **Note:** The positive zeros of J_ν are labelled λ_n, i.e., $J_\nu(\lambda_n) = 0$ for $n = 1, 2, 3, \ldots$.



Consequences

(i) If $\nu < 1/2$, between two zeros of $\alpha \cos x + \beta \sin x$, there is a root of $J_\nu(x)$, and the sequence of differences of roots of J_ν, $\{\lambda_{n+1} - \lambda_n\}$, is increasing and tends to π.

(ii) If $\nu = 1/2$, the difference between zeros of $J_\nu(x)$ is π.

(iii) If $\nu > 1/2$, between any two roots of $J_\nu(x)$, there is a root of $\alpha \cos x + \beta \sin x$, and the sequence of differences of roots of J_ν, $\{\lambda_{n+1} - \lambda_n\}$, is decreasing and tends to π.

- **Hint:** Use the Sturm comparison theorem for (i) and (iii).



The Self-Adjoint (Sturm–Liouville) Form

- The Bessel's equation is $x^2y'' + xy' + (\lambda^2x^2 - n^2)y = 0$.
- Divide through by x:

$$xy'' + y' + \left(\lambda^2 x - \frac{n^2}{x}\right) y = 0.$$

- The first two terms are exactly $\frac{d}{dx}\left[xy'\right] = xy'' + y'$, so

$$\boxed{\frac{d}{dx}\left[x\frac{dy}{dx}\right] + \left(\lambda^2 x - \frac{n^2}{x}\right) y = 0}$$

- This is Sturm–Liouville form $\frac{d}{dx}[p(x)y'] + [q(x) + \lambda^2 w(x)]y = 0$ with $p(x) = x$, $q(x) = -n^2/x$, and **weight function** $w(x) = x$. This weight function is used for the orthogonality of Bessel's functions over $[0, 1]$.
- The natural inner product is $\langle f, g\rangle = \int_0^1 x\, f(x)g(x)\, dx$.



Orthogonality of Bessel Functions

- **Theorem:** The Bessel functions J_ν $(\nu \geq 0)$ satisfy

$$\int_0^1 x J_\nu(\lambda_m x) J_\nu(\lambda_n x)\, dx = \frac{1}{2}\left(J_{\nu+1}(\lambda_n)\right)^2 \delta_{mn}, \tag{22}$$

  where λ_i are the positive zeros of J_ν, and $\delta_{mn} = 0$ for $m \neq n$ and $\delta_{mn} = 1$ for $m = n$.

- **Proof:** We know that $J_\nu(x)$ satisfies

$$y'' + \frac{1}{x}y' + \left(1 - \frac{\nu^2}{x^2}\right) y = 0. \tag{23}$$



Orthogonality of Bessel Functions (Contd.)

- Put $x = \lambda t$, $\lambda > 0$. Then

$$\frac{dJ_\nu(x)}{dx} = \frac{dJ_\nu(\lambda t)}{dt}\frac{dt}{dx} = \frac{1}{\lambda}\frac{dJ_\nu(\lambda t)}{dt}.$$

- Similarly, $\dfrac{d^2J_\nu(x)}{dx^2} = \dfrac{1}{\lambda^2}\dfrac{d^2J_\nu(\lambda t)}{dt^2}$.
- Substituting these changes in (23), we obtain

$$\frac{1}{\lambda^2}\frac{d^2J_\nu(\lambda t)}{dt^2} + \frac{1}{\lambda t}\frac{1}{\lambda}\frac{dJ_\nu(\lambda t)}{dt} + \left(1 - \frac{\nu^2}{\lambda^2 t^2}\right)J_\nu(\lambda t) = 0.$$

- Multiplying by λ^2 throughout:

$$\frac{d^2J_\nu(\lambda t)}{dt^2} + \frac{1}{t}\frac{dJ_\nu(\lambda t)}{dt} + \left(\lambda^2 - \frac{\nu^2}{t^2}\right)J_\nu(\lambda t) = 0.$$



Orthogonality of Bessel Functions (Contd.)

- If we write $u(x) = J_\nu(\lambda x)$ and $v(x) = J_\nu(\mu x)$, then u and v satisfy

$$u'' + \frac{1}{x}u' + \left(\lambda^2 - \frac{\nu^2}{x^2}\right)u = 0, \tag{24}$$

and

$$v'' + \frac{1}{x}v' + \left(\mu^2 - \frac{\nu^2}{x^2}\right)v = 0. \tag{25}$$

- Multiplying (24) by v and (25) by u, and subtracting, we obtain

$$(u''v - uv'') + \frac{1}{x}(u'v - uv') = (\mu^2 - \lambda^2)uv.$$



Orthogonality of Bessel Functions (Contd.)

- The last equation is equivalent to

$$\frac{d(u'v - uv')}{dx} + \frac{1}{x}(u'v - uv') = (\mu^2 - \lambda^2)uv,$$

$$x\frac{d(u'v - uv')}{dx} + (u'v - uv') = (\mu^2 - \lambda^2)xuv,$$

which can be written as

$$\frac{d}{dx}\Big[x(u'v - uv')\Big] = (\mu^2 - \lambda^2)\, xuv. \tag{26}$$

- Integrating from $x = 0$ to $x = 1$, we obtain

$$(\mu^2 - \lambda^2)\int_0^1 xuv\, dx = u'(1)v(1) - u(1)v'(1). \tag{27}$$

- Choose $\lambda = \lambda_m$ and $\mu = \lambda_n$, positive zeros of $J_\nu(x)$.
- Then $u(1) = J_\nu(\lambda_m) = 0$ and $v(1) = J_\nu(\lambda_n) = 0$.



Orthogonality of Bessel Functions (Contd.)

- Thus, from (27), we conclude that

$$(\lambda_n^2 - \lambda_m^2) \int_0^1 x J_\nu(\lambda_m x) J_\nu(\lambda_n x)\, dx = 0.$$

- The above relation gives that for $n \neq m$,

$$\int_0^1 x J_\nu(\lambda_m x) J_\nu(\lambda_n x)\, dx = 0.$$

- Replacing $u(x) = J_\nu(\lambda x)$, $v(x) = J_\nu(\mu x)$, $u'(x) = \lambda J'_\nu(\lambda x)$ and $v'(x) = \mu J'_\nu(\mu x)$ in (26), we have

$$\frac{d}{dx}\Big[x\big(\lambda J'_\nu(\lambda x) J_\nu(\mu x) - \mu J_\nu(\lambda x) J'_\nu(\mu x)\big)\Big] = (\mu^2 - \lambda^2)\, x J_\nu(\lambda x) J_\nu(\mu x).$$

- Differentiate this equation with respect to μ, then put $\mu = \lambda$, to obtain

$$2\lambda x J_\nu(\lambda x) J_\nu(\lambda x) = \frac{d}{dx}\Big[x\big(x\lambda J'_\nu(\lambda x) J'_\nu(\lambda x) - J_\nu(\lambda x) J'_\nu(\lambda x) - x\lambda J_\nu(\lambda x) J''_\nu(\lambda x)\big)\Big]$$



Orthogonality of Bessel Functions (Contd.)

- Note: use the product rule when differentiating $\mu J_\mu(x)$ and $\mu J'_\mu(x)$ w.r.t. μ; two terms cancel in the RHS when $\mu = \lambda$.
- Integrating both sides between $x = 0$ and $x = 1$:

$$2\lambda \int_0^1 x J_\nu^2(\lambda x)\, dx = \lambda \big(J'_\nu(\lambda)\big)^2 - J_\nu(\lambda) J'_\nu(\lambda) - \lambda J_\nu(\lambda) J''_\nu(\lambda).$$

- Dividing by 2λ throughout:

$$\int_0^1 x J_\nu^2(\lambda x)\, dx = \frac{1}{2} J'_\nu(\lambda)^2 - \frac{J_\nu(\lambda)}{2} \left(\frac{J'_\nu(\lambda)}{\lambda} + J''_\nu(\lambda) \right).$$

- If $\lambda = \lambda_n$, a positive zero of $J_\nu(x)$, the last two terms of the RHS vanish. Consequently,

$$\int_0^1 x J_\nu^2(\lambda_n x)\, dx = \frac{1}{2} \big(J'_\nu(\lambda_n)\big)^2. \tag{28}$$



Orthogonality of Bessel Functions (Contd.)

- From an earlier recurrence relation, it is known that

$$\left(x^{-\nu}J_\nu(x)\right)' = -x^{-\nu}J_{\nu+1}(x) \implies J'_\nu(x) - \frac{\nu}{x}J_\nu(x) = -J_{\nu+1}(x).$$

- Putting $x = \lambda_n$, a root of $J_\nu(x)$, gives

$$J'_\nu(\lambda_n) = -J_{\nu+1}(\lambda_n). \tag{29}$$

- Substituting (29) in (28), we get the second part of the orthogonality result:

$$\int_0^1 xJ_\nu^2(\lambda_n x)\,dx = \frac{1}{2}J_{\nu+1}^2(\lambda_n).$$



Fourier-Bessel Series

- **Theorem:** Suppose a function f is defined on $0 \leq x \leq 1$ and is square integrable. Then f has a Fourier-Bessel series expansion

$$f(x) \sim \sum_{n=1}^{\infty} a_n J_\nu(\lambda_n x), \tag{30}$$

where λ_n are the positive zeros of $J_\nu(x)$, and the coefficients a_n are given by

$$a_n = \frac{2}{J_{\nu+1}^2(\lambda_n)} \int_0^1 x f(x) J_\nu(\lambda_n x)\, dx. \tag{31}$$

- **Proof:** Assume that (30) is true, and let λ_m be a fixed positive zero of $J_\nu(x)$.
- Multiplying $xJ_\nu(\lambda_m x)$ on both sides of (30) and integrating between 0 and 1:



Fourier-Bessel Series (Contd.)

- $$\int_0^1 xf(x)J_\nu(\lambda_m x)\,dx = \int_0^1 \left(\sum_{n=1}^{\infty} a_n J_\nu(\lambda_n x)\right) xJ_\nu(\lambda_m x)\,dx$$

$$= \sum_{n=1}^{\infty} a_n \int_0^1 xJ_\nu(\lambda_n x)J_\nu(\lambda_m x)\,dx = a_m \frac{1}{2} J_{\nu+1}^2(\lambda_m).$$

- Therefore, we have

$$a_m = \frac{2}{J_{\nu+1}^2(\lambda_m)} \int_0^1 xf(x)J_\nu(\lambda_m x)\,dx.$$

- **Q.** Find the Fourier-Bessel series of $f(x) = 1$, $0 \leq x \leq 1$, in terms of Bessel functions of order zero.



Fourier-Bessel Series: Example

- **Solution:** Let $1 = f(x) \sim \sum_{n=1}^{\infty} a_n J_0(\lambda_n x)$, where λ_n are the positive zeros of $J_0(x)$, and

$$a_n = \frac{2}{J_1^2(\lambda_n)} \int_0^1 x f(x) J_0(\lambda_n x)\, dx = \frac{2}{J_1^2(\lambda_n)} \int_0^1 x J_0(\lambda_n x)\, dx.$$

- Ignoring the constant of integration in

$$\int x J_0(x)\, dx = x J_1(x)$$

and using the substitution $x = \lambda_n t$, we have

$$\int \lambda_n t J_0(\lambda_n t) \lambda_n\, dt = \lambda_n t J_1(\lambda_n t),$$

$$\implies \int t J_0(\lambda_n t)\, dt = \frac{1}{\lambda_n} t J_1(\lambda_n t).$$



Fourier-Bessel Series: Example (Contd.)

- Using the above relation to compute a_n:

$$a_n = \frac{2}{J_1^2(\lambda_n)} \int_0^1 x J_0(\lambda_n x)\, dx = \frac{2}{J_1^2(\lambda_n)} \left. \frac{x J_1(\lambda_n x)}{\lambda_n} \right|_0^1 = \frac{2}{\lambda_n J_1(\lambda_n)}.$$

- Therefore, the required Fourier-Bessel series is

$$1 \sim \sum_{n=1}^{\infty} \frac{2}{\lambda_n J_1(\lambda_n)} J_0(\lambda_n x).$$

- **Homework:** Find the Fourier-Bessel series of $f(x) = x$, $0 \leq x \leq 1$, in terms of Bessel functions of order one.
- **Answer:**

$$x \sim \sum_{n=1}^{\infty} \frac{2}{\lambda_n J_2(\lambda_n)} J_1(\lambda_n x).$$



Fourier–Bessel series for $f(x) = x^2$

- **Problem.** Expand $f(x) = x^2$ in a Fourier–Bessel series of order $\nu = 1$ and of order $\nu = 2$.
- In both cases $xf(x) = x^3$, so

$$\int_0^1 x^3 J_\nu(\lambda_n x)\, dx = \frac{1}{\lambda_n^4} \int_0^{\lambda_n} t^3 J_\nu(t)\, dt \qquad (t = \lambda_n x).$$

- Here λ_n are the positive zeros of J_1, so $J_1(\lambda_n) = 0$.
- Recurrence at $\nu = 1$, $x = \lambda_n$:

$$J_0(\lambda_n) + J_2(\lambda_n) = \frac{2}{\lambda_n} J_1(\lambda_n) = 0 \implies J_2(\lambda_n) = -J_0(\lambda_n).$$

- Formula (31) becomes

$$a_n = \frac{2}{J_0^2(\lambda_n)} \cdot \frac{1}{\lambda_n^4}\, I(\lambda_n), \qquad I(\lambda) = \int_0^{\lambda} t^3 J_1(t)\, dt.$$



$\nu = 1$: Computing $I(\lambda)$

- Use $J_1 = -J_0'$ and integrate by parts:

$$I(\lambda) = -\lambda^3 J_0(\lambda) + 3\int_0^\lambda t^2 J_0(t)\,dt.$$

- Since $tJ_0 = (tJ_1)'$:

$$\int_0^\lambda t^2 J_0\,dt = \lambda^2 J_1(\lambda) - \int_0^\lambda tJ_1\,dt.$$

- Again with $J_1 = -J_0'$:

$$\int_0^\lambda tJ_1\,dt = -\lambda J_0(\lambda) + \int_0^\lambda J_0(t)\,dt.$$

- Hence

$$I(\lambda) = -\lambda^3 J_0(\lambda) + 3\lambda^2 J_1(\lambda) + 3\lambda J_0(\lambda) - 3\int_0^\lambda J_0(t)\,dt.$$

- At $\lambda = \lambda_n$ the term $3\lambda_n^2 J_1(\lambda_n)$ vanishes.



$\nu = 1$: Coefficients and Series

- Substituting in (31):

$$a_n = \frac{2(3-\lambda_n^2)}{\lambda_n^3 J_0(\lambda_n)} - \frac{6}{\lambda_n^4 J_0^2(\lambda_n)} \int_0^{\lambda_n} J_0(t)\, dt.$$

- Therefore, by (30),

$$x^2 \sim \sum_{n=1}^{\infty} \left[\frac{2(3-\lambda_n^2)}{\lambda_n^3 J_0(\lambda_n)} - \frac{6\int_0^{\lambda_n} J_0(t)\, dt}{\lambda_n^4 J_0^2(\lambda_n)} \right] J_1(\lambda_n x).$$

- Numerically ($\lambda_1 \approx 3.8317$, $\lambda_2 \approx 7.0156$, $\lambda_3 \approx 10.1735$):

$$a_1 \approx 0.8437, \quad a_2 \approx -0.9184, \quad a_3 \approx 0.7553.$$



$\nu = 2$: Computing the Integral

- Here λ_n are the positive zeros of J_2, so $J_2(\lambda_n) = 0$.
- Since $\dfrac{d}{dt}\left[t^3 J_3(t)\right] = t^3 J_2(t)$,

$$\int_0^{\lambda} t^3 J_2(t)\, dt = \lambda^3 J_3(\lambda).$$

- Therefore

$$\int_0^1 x^3 J_2(\lambda_n x)\, dx = \frac{1}{\lambda_n^4}\, \lambda_n^3 J_3(\lambda_n) = \frac{J_3(\lambda_n)}{\lambda_n}.$$



$\nu = 2$: Coefficients and Series

- Substituting in (31) with $\nu = 2$:

$$a_n = \frac{2}{J_3^2(\lambda_n)} \cdot \frac{J_3(\lambda_n)}{\lambda_n} = \frac{2}{\lambda_n J_3(\lambda_n)}.$$

- Therefore, by (30),

$$\boxed{x^2 \sim 2 \sum_{n=1}^{\infty} \frac{J_2(\lambda_n x)}{\lambda_n J_3(\lambda_n)}}$$

- Numerically ($\lambda_1 \approx 5.1356$, $\lambda_2 \approx 8.4172$, $\lambda_3 \approx 11.6198$):

$$a_1 \approx 1.1465, \quad a_2 \approx -0.8755, \quad a_3 \approx 0.7405.$$



Convergence of Fourier-Bessel Series

- **Convergence in Norm:** The Fourier-Bessel series converges to $f(x)$ in the weighted norm ($w_x = x$), i.e. $\|f(x) - \sum_{n=1}^{m} a_n J_\nu(\lambda_n x)\| \to 0$ as $m \to \infty$.
- For pointwise convergence, we have the **Bessel expansion theorem**: suppose f and f' are piecewise continuous on $0 \le x \le 1$. Then for $0 < x < 1$,

$$\sum_{n=1}^{\infty} a_n J_\nu(\lambda_n x) = \begin{cases} f(x), & \text{where } f \text{ is continuous} \\ \dfrac{f(x^-) + f(x^+)}{2}, & \text{where } f \text{ is discontinuous} \end{cases}$$

- At $x = 0$, the series converges to 0 if $\nu > 0$.
- At $x = 0$, the series converges to $f(0^+)$ if $\nu = 0$.
- At $x = 1$, the series always converges to 0.



Parametric Form of Bessel's Equation

- **Result:** Let $\nu \geq 0$ and $a > 0$. Let λ_n be the positive zero of $J_\nu(x)$ for $n \in \mathbb{N}$. Then $J_\nu\left(\frac{\lambda_n x}{a}\right)$ are solutions of the parametric form of Bessel's equation of order ν:

$$x^2 y'' + xy' + (p^2 x^2 - \nu^2) y = 0 \tag{32}$$

with boundary conditions $y(0)$ finite and $y(a) = 0$, where $p = p_n = \lambda_n / a$; apart from scalar multiples, these are the only solutions with these properties.

- (Hint: Use $u = px$ in (32), take $y(x) = y(u/p) = Y(u)$ and apply the boundary conditions.)
- These solutions satisfy orthogonality properties over $[0, a]$ with respect to the weight function $w(x) = x$. Therefore,

$$\int_0^a x J_\nu(p_m x) J_\nu(p_n x)\, dx = \frac{a^2}{2} \left(J_{\nu+1}(p_n)\right)^2 \delta_{mn}. \tag{33}$$

